

Syllabary spelling

One of the common keys to breaking into some codes and ciphers is identifying and exploiting syllabary spelling. This essentially includes identification of instances where the same word is spelled in different ways by combining the syllables and letters in different combinations each time. This method is fairly easy if applied to the appropriate code, since most syllabic and letter clusters tend to stick together. By identifying these repeating patterns, one can write down the message in syllabary language, rearrange it on the basis of the sounds of the syllables and the letters and then decode the plaintext message from the rearranged syllabary message. For example, the last letters in words ending with the ‘-ur’ sound are always ‘re’, such as fracture, departure, capture, etc. We can use similar rules that apply to pronunciation of the English language to easily decrypt ciphertext.

Cryptanalysis today

As mentioned earlier, the abovementioned methods are elementary at best, and are unlikely to be of any use in the “real world” of cryptography today. Despite the successful use of computation to break cryptographic systems during and since World War II, the improvement in technology and knowledge has also vastly improved the complexity of new methods of cryptography. On the whole, modern cryptography has become much more resistant to cryptanalysis than the systems of the past.

However, cryptanalysis is far from dead. It has simply had to evolve. Traditional methods of cryptanalysis have given way to new techniques including interception, bugging, side channel attacks, quantum computers, etc. The effectiveness of the cryptanalysis techniques used by government and law enforcement agencies is a tightly guarded secret; but there have been some major breakthroughs recently against both academic (purely theoretical) as well as practical cryptographic systems. Thus, while modern ciphers and codes may be far advanced as compared to even the best of the codes from the past, such as the Enigma Cipher from World War II, Cryptanalysis remains very much active and thriving.

Conclusion

Cryptanalysis is one of the fields that appears to be shrouded in mystery. Commonly associated with spies and the stuff of detective novels, it is in fact a purely logical field that relies frequently on little more than common